

Extra practice problems

1. (2022/23 Sem 1) Let $X_1, \dots, X_n$ be a random sample with density

$$f(x; \lambda) = \lambda e^x \exp(-\lambda e^x),$$

where $\lambda > 0$.

- (a) Find a one-dimensional sufficient statistic for λ .
- (b) Consider testing $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda = \lambda_1$, where λ_0 and λ_1 are positive constants with $\lambda_1 < \lambda_0$. Show that the most powerful test of a given size has rejection region

$$\left\{ X_1, \dots, X_n : \sum_{i=1}^n e^{X_i} > c \right\}$$

for some constant c .

- (c) Find the value of c in the rejection region in (b), such that the test has size α ($0 < \alpha < 1$). Express c in terms of α and quantiles of a chi-square distribution. (Hint: e^{X_i} follows the exponential distribution with rate λ for $i = 1, \dots, n$.)
- (d) Argue that the test in (b) with the value of c you obtained in (c) is the uniformly most powerful size α test for testing $H_0 : \lambda \geq \lambda_0$ versus $H_1 : \lambda < \lambda_0$.
2. (2023/24 Sem 1) Let X be one observation from the Beta(1, θ) distribution with density

$$f(x; \theta) = \theta(1-x)^{\theta-1} \quad \text{for } 0 < x < 1,$$

where $\theta > 0$. Consider testing $H_0 : \theta = 1$ versus $H_1 : \theta = 2$.

- (a) Find the most powerful size α test.
- (b) Find the size of the test that rejects H_0 if and only if $X < 1/4$.
- (c) Consider the test in (b). If $\theta = 2$, then what is the rejection probability of the test?
3. (2023/24 Sem 1) Let $X_1, \dots, X_n$ be a random sample from the Poisson distribution with mean θ , where the probability mass function is

$$f(x | \theta) = \frac{e^{-\theta} \theta^x}{x!} \quad \text{for } x = 0, 1, 2, \dots,$$

and $\theta > 0$. Adopt a Bayesian paradigm, and set the following prior distribution for θ :

$$\pi(\theta) = 4\theta^2 e^{-2\theta} \quad \text{for } \theta > 0.$$

- (a) State the posterior distribution of θ . You can simply give the name and parameter values of the distribution (in terms of $X_1, \dots, X_n$ and n).
- (b) Give the Bayes estimator of θ under the squared error loss.
- (c) Give the Bayes estimator of $1/\theta$ under the squared error loss.
- (d) Derive a $(1 - \alpha)$ credible interval for θ . Give the interval in terms of the data, α , and quantiles of a chi-square distribution.
4. (2024/25 Sem 1) Let $X_1, \dots, X_n$ be a random sample from the $N(\mu, 1)$ distribution, where μ is an unknown parameter. We are interested in testing $H_0 : \mu = 0$ versus $H_1 : \mu = 0.5$, and we consider the test that rejects H_0 if and only if $n^{1/2}\bar{X} > c$ for some constant c , where $\bar{X} = n^{-1} \sum_{i=1}^n X_i$. Find n and c such that the test has type I error probability of 0.1 and power of 0.9. The following quantiles of the standard normal distribution may be useful: $z_{0.2} = 0.842$, $z_{0.1} = 1.282$, $z_{0.05} = 1.645$, $z_{0.025} = 1.960$.
5. (2024/25 Sem 1) Let $X_1, \dots, X_n$ be a random sample, with density

$$f(x; \theta) = 2\theta x e^{-\theta x^2} \quad \text{for } x > 0, \text{ where } \theta > 0.$$

- (a) Consider testing $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ for some $\theta_0 > \theta_1 > 0$. Show that the most powerful test takes the form of

$$\text{reject } H_0 \text{ if and only if } \sum_{i=1}^n X_i^2 > c \quad \text{for some constant } c.$$

- (b) Express the power function of the test in (a) in terms of the distribution function of a chi-square random variable. You can use the fact that $\sum_{i=1}^n X_i^2$ follows the $\text{Gamma}(n, \theta)$ distribution.
- (c) Argue that the test in (a) is the uniformly most powerful test for $H_0 : \theta \geq \theta_0$ against $H_1 : \theta < \theta_0$ among tests of the same type I error probability.
6. (2024/25 Sem 1) Let $X_1, \dots, X_n$ be a random sample, with density

$$f(x; \theta) = \frac{\theta}{x^2} \quad \text{for } x > \theta,$$

where $\theta > 0$. Let $X_{(1)} = \min_{i=1, \dots, n} X_i$.

- (a) Show that $P(X_{(1)} > x) = (\theta/x)^n$ for $x > \theta$.
- (b) Find the value of t such that

$$P\left(X_{(1)} > \frac{\theta}{t}\right) = 1 - \alpha$$

for a given $0 < \alpha < 1$. Then, find a one-sided upper $(1 - \alpha)$ confidence limit for θ .